
Synthetic American Option Pricing via Jump-HMM-Driven Heston Implied Volatility

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Abstract

Generating realistic synthetic option prices requires implied volatility as an input, yet implied volatility is itself derived from observed option prices, creating a circular dependency that limits synthetic data availability for machine-learning and risk-analysis applications. We broke this circularity by developing a pipeline in which implied volatility emerged as an output of a structural model of equity returns rather than as an exogenous input. A Jump Hidden Markov Model produced multi-asset price paths with realistic stylized facts and cross-asset tail dependence, a modified Heston stochastic variance process whose mean-reversion target depended on regime state, days to expiration, moneyness, and an aggregate market-mood indicator converted those paths into implied-volatility paths, and a Cox-Ross-Rubinstein binomial tree priced American options from the resulting surface with early exercise. A central design choice was initializing the variance process at its mean-reversion target for each strike-expiration pair, so that the smile, skew, and term structure emerged automatically without external calibration. We calibrated the shape function through a hierarchy of representations spanning a parametric baseline, a globally shared neural surrogate, and a sector-specific neural surrogate fit to a 31-ticker, six-sector option ladder. A temporal holdout on a multi-day capture isolated scheduled corporate events as the dominant source of test-time generalization error, and calendar-derived earnings-distance and same-sector peer-coupling features fed to the shape network recovered the anticipatory portion of that signal. We then exercised the calibrated framework as a synthetic-data generator on real 30-delta LLY put and call contracts pulled from a single capture day, forward-simulating price paths under the JumpHMM and the per-ticker shape network and recovering path-conditional implied volatility, finite-difference Leisen-Reimer American Greeks, and terminal short-premium profit and loss from one coherent simulation. The framework was implemented as an open-source Julia package designed to produce synthetic option-market data for downstream applications.

1 Introduction

Pricing American-style options requires implied volatility (IV) as an input, yet IV itself is an empirical quantity extracted from observed option prices [Black and Scholes \[1973\]](#). This circularity poses a practical problem for synthetic data generation: existing approaches either take IV from the market and feed it into a pricing model, or generate price paths without a consistent IV surface. Parametric IV models such as SABR [Hagan et al. \[2002\]](#) and the SVI parameterization [Gatheral \[2004\]](#) treat the IV surface as an object to be fitted to market quotes, not generated from first principles; they produce smooth interpolations of observed smiles but cannot generate new IV dynamics for unobserved market conditions. Local volatility models [Dupire \[1994\]](#), [Derman and Kani \[1994\]](#)

recover a deterministic volatility surface consistent with a single cross-section of option prices, yet the resulting surface is static and cannot produce the stochastic IV fluctuations observed in real markets. Stochastic volatility models such as Heston [Heston \[1993\]](#) generate realistic variance paths, but calibrating them requires observed option prices or IV as input, reintroducing the very circularity that the synthetic generator must avoid. The result is a gap in the literature: no existing framework generates self-consistent IV surfaces as an emergent output of a structural model of equity returns, without requiring observed option data as a calibration target during simulation.

This gap has practical consequences for a growing set of downstream applications. Empirical studies of IV surface dynamics [Cont and da Fonseca \[2002\]](#) have established that implied volatility exhibits rich cross-sectional and temporal structure, including smile curvature, skew, and term-structure effects that interact with market regimes. Reproducing this structure in synthetic data is essential for training machine-learning models that must generalize across market conditions, for backtesting hedging strategies such as deep hedging [Buehler et al. \[2019\]](#), and for conducting scenario-based risk analysis that requires consistent option prices across strikes and expirations. Historical re-sampling preserves realized distributional features but cannot produce out-of-sample tail scenarios consistent with a structural model of returns; generative approaches based on neural networks have demonstrated promise for equity price paths but do not directly produce self-consistent option price and IV pairs. The scarcity of realistic synthetic option data therefore remains a binding constraint on progress in computational finance.

In this study, we addressed this gap by developing a framework in which IV was an *output* of a structural model of equity returns, not an input. The pipeline coupled a Jump Hidden Markov Model (JumpHMM) [Alswaidan and Varner \[2025\]](#) that produced multi-asset price paths with realistic heavy tails, negligible linear autocorrelation, and persistent volatility clustering [Cont \[2001\]](#), with cross-asset dependence imposed via a Student- t copula [Demarta and McNeil \[2005\]](#); a modified Heston stochastic variance process [Heston \[1993\]](#) that converted those price paths into IV paths, where the mean-reversion target θ was a function of the HMM regime state, days to expiration (DTE), money-ness, and an aggregate market mood indicator; and a Cox-Ross-Rubinstein (CRR) binomial tree [Cox et al. \[1979\]](#) that converted the pre-computed IV into American option prices with early exercise. The central innovation was a hybrid θ -function that linked the Heston mean-reversion target to the JumpHMM regime state, producing regime-dependent volatility dynamics. An equilibrium initialization $v_0 = \theta(t = 0)$ gave each (ticker, strike, DTE) triple its own initial IV, so the smile, skew, and term structure emerged automatically without external IV data. Within this pipeline, the shape function that controlled the smile and term structure of θ admitted a family of representations, and the modeling choice at each level of the family traded off model capacity for sharing across tickers. We therefore calibrated the shape function through a hierarchy of increasingly expressive forms: a parsimonious five-parameter functional form on single-ticker and cross-ticker semiconductor option chains, a globally shared neural surrogate that replaced the analytic form with a data-driven representation, and a sector-specific neural surrogate that let within-sector shapes specialize. The same decomposition supports a further relaxation to per-ticker networks once per-name sample sizes grow, without changing the downstream pipeline. We demonstrated this progression on a 31-ticker, two-day, six-sector option ladder and on an eight-day temporal-holdout extension, the latter of which isolated scheduled earnings events as the dominant source of the temporal generalization gap and motivated calendar-derived earnings-distance and same-sector peer-distance features that recovered a portion of that gap. The framework also produced realistic forward IV paths and American option prices with an endogenous crash-protection premium. The framework is implemented as an open-source Julia package designed to serve as a synthetic option market generator for downstream applications.

2 Background

The framework built on three established models, each contributing a distinct component of the synthetic option generation pipeline: the JumpHMM produced realistic multi-asset price paths, the Heston stochastic variance process provided a mechanism for converting price dynamics into implied volatility, and the CRR binomial tree translated IV into American option prices with early exercise.

2.1 Jump Hidden Markov Model

The Jump Hidden Markov Model (JumpHMM) [Alswaidan and Varner \[2025\]](#) was developed for generating synthetic equity time series that simultaneously preserved three stylized facts of financial returns: heavy tails, negligible linear autocorrelation, and persistent volatility clustering [Cont \[2001\]](#). The model discretized continuous excess growth rates into N states via equal-probability quantile bins of a fitted Laplace distribution. The hidden state $S_t \in \{1, \dots, N\}$ at each timestep governed the distribution of the excess growth rate G_t ; each state k emitted observations from a location-scale Student- t distribution: $G_t | S_t = k \sim \mu_k + \sigma_k \cdot t_\nu$, where μ_k and σ_k were learned per-state location and scale parameters and ν was the shared degrees-of-freedom parameter. At each timestep, with probability ϵ a Poisson(λ)-distributed jump forced the chain into tail states (bottom or top N_{tail} states), creating sustained dwell periods in extreme-volatility regimes that produced the volatility clustering characteristic of empirical return series. For multi-asset generation, cross-asset dependence was imposed via a Student- t copula [Demarta and McNeil \[2005\]](#) that preserved each asset’s marginal distribution while introducing realistic tail dependence; this copula structure ensured that joint tail events, in which multiple assets simultaneously occupied extreme states, occurred more frequently than a Gaussian copula would imply.

2.2 Heston stochastic variance model

The Heston model [Heston \[1993\]](#) specified the instantaneous variance v of the underlying asset as a mean-reverting square-root process:

$$dv = \kappa(\theta - v) dt + \sigma_v \sqrt{v} dW_v \quad (1)$$

where κ was the mean-reversion speed controlling how quickly the variance returned to its long-run level, θ was the long-run variance target, and σ_v was the vol-of-vol governing the amplitude of stochastic fluctuations in v . In the standard formulation, the asset price and variance processes were coupled through a correlation ρ between their driving Brownian motions, producing the leverage effect in which negative returns coincided with rising volatility. The key property for our purposes was mean-reversion: the variance process was pulled toward θ , so by controlling θ we could control the equilibrium level of implied volatility. In the standard model θ was a constant; our extension made θ time-varying and state-dependent, linking it to the JumpHMM regime state, contract characteristics, and market conditions.

2.3 CRR binomial tree

The CRR model [Cox et al. \[1979\]](#) priced options on a recombining binomial lattice with up/down factors $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$ and risk-neutral probability $p = (e^{r\Delta t} - d)/(u - d)$. American options were priced by backward induction with an early-exercise check at each node, making the CRR tree a standard tool for contracts where closed-form solutions do not exist. The CRR tree was a natural choice when IV was provided exogenously, as it required only a single volatility input per contract and handled early exercise without the need for Monte Carlo simulation or PDE solvers. Convergence to the continuous-time Black-Scholes price was guaranteed as the number of tree steps increased, and the computational cost scaled linearly in the number of steps, making the tree efficient for pricing large cross-sections of contracts across multiple strikes and expirations.

2.4 Neural representations of the implied volatility surface

Parametric IV models such as SABR [Hagan et al. \[2002\]](#) and SVI [Gatheral \[2004\]](#) expressed the smile and term structure through low-dimensional closed-form functions, chosen to interpolate observed quotes while respecting no-arbitrage constraints. These forms were parsimonious and interpretable but inherited their expressive capacity from the chosen functional family, and a single global parameterization across a broad universe of tickers could not simultaneously accommodate sector-specific skew patterns and smile curvatures. A complementary line of work [Horvath et al. \[2021\]](#), [Bayer and Stemper \[2019\]](#) replaced the analytic form with a neural network that took $(\ln \tau, \ln m)$ or a similar low-dimensional set of contract and model features as input and returned an IV or variance output; the network provided a flexible universal approximator for the shape of the surface while the downstream pricing pipeline remained unchanged. Neural surrogates of this kind have been used both as fast pricing approximators and as calibration targets in deep calibration schemes, and they

admit the same multiplicative decomposition between a per-ticker level and a shared shape that the parametric forms use, making them a drop-in replacement for the functional form inside a generative pipeline.

3 Methods

3.1 Pipeline overview

The generator produced synthetic American option prices through a three-stage pipeline:

$$\begin{aligned} \underbrace{\text{JumpHMM}(\text{prices})}_{\text{Stage 1}} &\rightarrow \{S_t, s_t, \text{jumps}\} \rightarrow \underbrace{dv \text{ process}}_{\text{Stage 2}} \rightarrow \sigma_{\text{imp}} = \sqrt{v_t} \\ &\rightarrow \underbrace{\text{CRR}(S_t, \sigma_{\text{imp}}, K, \text{DTE}, r_f)}_{\text{Stage 3}} \rightarrow P_{\text{american}} \end{aligned} \quad (2)$$

This ordering was dictated by data flow: the JumpHMM produced the price paths and regime-state sequences that the Heston process required as inputs, and the Heston process produced the IV values that the CRR tree required for pricing. Each stage was self-contained, so the pipeline could be run forward without iteration or feedback from downstream stages.

3.2 Modified Heston variance process

We modified the standard Heston process (1) by making the mean-reversion target time-varying:

$$dv = \kappa(\theta(t) - v) dt + \sigma_v \sqrt{v} dW_v \quad (3)$$

The mean-reversion target decomposed into a per-ticker base level, an aggregate market mood modulator, and a contract-specific smile and term-structure adjustment that included a scheduled-event indicator:

$$\theta(i, t) = \theta_{i, s_t} \cdot (1 + \gamma \cdot M_t) \cdot \psi(\text{DTE}_t, K/S_t, e_{i,t}, e_{i,t}^{\text{peer}}) \quad (4)$$

where θ_{i, s_t} was a per-ticker, regime-dependent variance level (one value per ticker i per HMM state s_t); $\gamma \geq 0$ was the market mood sensitivity controlling how strongly aggregate stress elevated IV; $M_t \in [0, 1]$ was the aggregate market mood; and ψ was the smile, term-structure, and event-shape adjustment with two earnings-event inputs $e_{i,t}$ (signed days from t to ticker i 's nearest earnings print) and $e_{i,t}^{\text{peer}}$ (the minimum of $|e_{j,t}|$ over same-sector equity peers $j \neq i$). The regime-dependent base θ_{i, s_t} linked the Heston process directly to the JumpHMM state sequence for forward simulation: when the HMM transitioned into a tail state (via a Poisson jump or an extreme Markov transition), θ_{i, s_t} rose, elevating the IV target. This mechanism produced the leverage effect as an emergent property of the return dynamics rather than as a separate model component: large negative returns drove the HMM into low-numbered tail states with $p_{\text{neg}} \approx 0.52$, which in turn elevated θ . For the IV-surface calibration that follows, we restricted (4) to the operational form

$$\theta_{\text{cal}}(i, t) = \theta_i \cdot \psi(\text{DTE}_t, K/S_t, e_{i,t}, e_{i,t}^{\text{peer}}) \quad (5)$$

in which $\gamma = 0$ (the aggregate mood multiplier is reserved for forward simulation with simulated state paths) and the per-state level θ_{i, s_t} collapses to a single per-ticker level θ_i (since the calibration corpus consisted of observation-day IV surfaces rather than state-conditioned histories). Both restrictions were transparent to ψ , so the shape function carried over without modification when the full state-and-mood mechanism was reactivated for simulation.

The function ψ captured the shape of the IV surface with five parameters:

$$\psi(\tau, m) = \exp\left(\beta_1 \ln \tau + \beta_2 \ln m + \beta_3 \ln \tau \ln m + \beta_4 (\ln m)^2 + \beta_5 (\ln \tau)^2\right) \quad (6)$$

where $\tau = \max(\text{DTE}, 1)$ and $m = K/S$. The parameter β_1 controlled the linear term-structure decay; β_2 controlled the skew (negative values produced put skew, with OTM puts having higher IV than OTM calls); β_3 captured the DTE-skew interaction (the skew flattened at longer maturities); β_4 controlled the smile curvature; and β_5 controlled the DTE curvature, capturing the U-shaped ATM term structure where short-dated and long-dated IV were both elevated relative to intermediate maturities.

Rather than specifying a separate initial variance v_0 , we initialized the variance process at its mean-reversion target:

$$v_0 = \theta(s_0, \text{DTE}, K/S_0, M_0) \quad (7)$$

This ensured that each (ticker, strike, DTE) triple started at its own equilibrium implied volatility. The initial IV surface inherited the smile, skew, and term structure encoded in ψ without requiring any external IV data. Different contracts on the same underlying received different v_0 values, and the variance process immediately began mean-reverting around its local $\theta(t)$. The value of this design choice lay in eliminating v_0 as a free parameter: the model did not require a separate calibration of the initial IV surface, as it was determined by construction from θ .

The market mood M_t was defined as the fraction of tickers currently occupying tail HMM states, where N_{tail} denoted the number of extreme states at each end of the state space (inherited from the JumpHMM fit):

$$M_t = \frac{1}{N_{\text{tickers}}} \sum_{i=1}^{N_{\text{tickers}}} \mathbf{1}\{s_t^{(i)} \leq N_{\text{tail}} \text{ or } s_t^{(i)} > N - N_{\text{tail}}\} \quad (8)$$

This signal was fully endogenous to the JumpHMM copula model; it required no external data (e.g., VIX) and introduced no circularity. When multiple tickers simultaneously occupied tail states, the mood rose, elevating θ and thus IV across all contracts. This mechanism produced correlated IV spikes during market stress events by construction.

The variance process was discretized via Euler-Maruyama with a reflecting boundary at zero and a daily time step $\Delta t = 1/252$:

$$v_{t+1} = \left| v_t + \kappa(\theta_t - v_t)\Delta t + \sigma_v \sqrt{\max(v_t, 0)} \sqrt{\Delta t} Z_t \right|, \quad Z_t \sim \mathcal{N}(0, 1) \quad (9)$$

The reflecting boundary allowed θ to vary freely across HMM states without requiring the Feller condition $2\kappa\theta > \sigma_v^2$ to hold at every timestep Andersen [2008]. This was a practical necessity, as tail states could drive θ to values where the Feller condition would otherwise be violated. American option prices were computed from the resulting IV using a CRR binomial tree with 200 steps per contract, which provided sufficient convergence for the DTE range considered.

3.3 Neural surrogate for ψ

The parametric form of ψ in equation (6) used five shape parameters shared across all contracts on a given ticker. When the model was applied to a broad universe spanning multiple sectors, this global shape became a bottleneck: the semiconductor smile geometry differed systematically from mega-cap ETFs, healthcare names carried idiosyncratic event risk, and energy tickers inherited commodity-driven skew patterns not expressible by a five-term log-polynomial. We therefore replaced the parametric ψ with a neural surrogate Horvath et al. [2021], keeping the multiplicative decomposition $\sigma_{\text{model}}^2 = \theta_{\text{ticker}} \cdot \psi$ but allowing the shape to be data-driven and adding two scheduled-event inputs (e , e^{peer}) that captured pre-earnings IV expansion and same-sector earnings spillover:

$$\psi_{\text{NN}}(\ln \tau, \ln m, e, e^{\text{peer}}) = \exp(\text{NN}(\tilde{z}_\tau, \tilde{z}_m, \tilde{z}_e, \tilde{z}_{e^{\text{peer}}})) \quad (10)$$

where each input was standardized to its training-set mean and standard deviation, $\tilde{z}_x = (x - \mu_x)/\sigma_x$. The earnings inputs were the signed clipped days-to-earnings for the contract's ticker, $e = \text{clip}(\Delta_{\text{earn}}^{(i)}, -30, 30)$, and the minimum absolute days-to-earnings over same-sector equity peers excluding self, $e^{\text{peer}} = \min_{j \neq i, c(j)=c(i)} |\Delta_{\text{earn}}^{(j)}|$ (clipped to 30). The peer feature carried the sector-coupling signal: a ticker j in the same sector printing on day t contributed $e^{\text{peer}} = 0$ for all of i 's contracts that day, which let the network learn the IV-spillover shape characteristic of the sector. For ETFs (no own earnings), e was set equal to e^{peer} taken over the full equity universe. The network produced $\ln \psi$ so that $\psi > 0$ by construction, preserving the log-Gaussian interpretation of the smile surface used in the parametric form. Each sector network was a two-hidden-layer multilayer perceptron with tanh activations: $4 \rightarrow 16 \rightarrow 16 \rightarrow 1$ (369 parameters) for groups with ≥ 2000 observations and $4 \rightarrow 8 \rightarrow 8 \rightarrow 1$ (121 parameters) for smaller groups, a width chosen to keep the observations-to-parameters ratio above 30 across all sector fits.

The decomposition defined a natural hierarchy of sharing. At one extreme, a single global ψ_{NN} shared across all tickers assumed that smile geometry was universal and left only the variance level

θ_{ticker} to distinguish names. At the other extreme, a per-ticker ψ_{NN} recovered ticker-specific shape but required enough data per name to resolve the surface without overfitting. We adopted a sector-specific intermediate: one $\psi_{\text{NN}}^{(c)}$ per sector c , with all tickers within a sector sharing shape but each ticker retaining its own θ_{ticker} . This choice reflected the empirical observation that within-sector tickers shared dominant risk factors (technology names tracked semiconductor cycles, healthcare names tracked FDA and trial catalysts, financials tracked rate expectations), so pooling shape across a sector provided a favorable bias-variance tradeoff at the current data volume. As per-ticker sample sizes grew, the same architecture supported relaxation to per-ticker networks without changing the downstream pipeline.

Training minimized the mean squared IV error jointly over the network weights and the per-ticker log-variance levels $\{\ln \theta_{\text{ticker}}\}$:

$$\min_{w, \{\ln \theta_t\}} \frac{1}{n} \sum_{i=1}^n \left(\sqrt{\theta_{t(i)} \cdot \psi_{\text{NN}}(\ln \tau_i, \ln m_i, e_i, e_i^{\text{peer}}; w)} - \text{IV}_{\text{market}, i} \right)^2 \quad (11)$$

where $t(i)$ denoted the ticker of the i th observation. The log-variance levels were initialized from each ticker’s mean squared market IV and then updated jointly with the network weights under a single Adam optimizer, which coupled the level and shape fits and avoided the chicken-and-egg problem of fitting one conditional on the other. The learning rate followed a step schedule (1e-3 $\rightarrow$ 5e-4 at epoch 500 $\rightarrow$ 2e-4 at 1000 $\rightarrow$ 1e-4 at 1500), and training ran up to 2000 epochs with a best-checkpoint patience of 200 epochs on the training loss. Earnings dates were sourced from the public Yahoo Finance calendar and joined to each contract by ticker and observation date.

4 Calibration and Validation

4.1 Parametric baseline on NVDA

Before replacing the parametric form of ψ with a neural surrogate, we established the baseline capacity of the five-parameter analytic form on a well-characterized single-ticker dataset. We collected NVDA call and put chains at seven expirations on the same observation date (April 6, 2026): 0, 2, 4, 7, 14, 25, and 46 DTE, with NVDA trading near \$176.70, yielding 73 observations after filtering to moneyness $K/S \in [0.90, 1.10]$ with positive volume. The parametric ψ of equation (6) was fitted by minimizing mean squared IV error across all seven expirations simultaneously. The calibrated form captured the full ATM term structure, including the U-shape driven by gamma compression at short maturities and uncertainty premia at longer horizons, achieving an RMSE of 1.15% IV across the 2 to 46 DTE range. The quadratic DTE term β_5 was the critical degree of freedom: without it the linear $\beta_1 \ln \tau$ term could not bend upward at long maturities, and the U-shape was unreachable; with it the parametric form reproduced the trough near DTE ≈ 8 at 32.5% IV and the subsequent rise to 41% at 46 DTE that characterized the market data. This 1% RMSE on a single well-sampled ticker established the upper bound of what the parametric form could deliver under favorable conditions. Additional parametric calibrations, including a single-DTE cross-ticker validation on four semiconductor names, the per-parameter interpretation table, and full residual diagnostics by moneyness and DTE, are reported in the supplement. Whether the same shape transferred across a broader universe, and how much a more flexible representation could improve the fit when the universe was heterogeneous, motivated the calibration on the full 31-ticker ladder below.

4.2 Sector-specific neural calibration on the 31-ticker ladder

The parametric baseline showed that the five-parameter ψ could reach near-1% RMSE on a single well-sampled ticker, but left open whether the same shape transferred across sectors with heterogeneous risk profiles. To answer this directly, we collected full option ladders for 31 tickers spanning six sectors on two consecutive trading days (2026-04-14 and 2026-04-15), producing 35,961 observations after filtering to moneyness $K/S \in [0.80, 1.20]$, non-zero bids, and $\text{IV} \in (0.01, 2.0)$. The sector partition followed conventional GICS-style groupings (Table 1): technology (10 tickers), healthcare (8), financials (4), energy (3), retail (3), and broad-market ETFs (3). This corpus was two orders of magnitude larger than the NVDA multi-expiration dataset and, more importantly, contained systematic cross-sector variation: mega-cap ETFs exhibited shallow smiles with pronounced term structure, technology names carried elevated near-term skew from earnings catalysts, healthcare names showed idiosyncratic bumps around trial and FDA dates, and commodity-linked energy

tickers displayed skew patterns shaped by supply shocks rather than equity crash risk. Against this partition, we quantified the value of relaxing the global-shape assumption by fitting three nested models on the pooled ladder using a common loss (mean squared IV error) and a common level parameterization (one θ_{ticker} per ticker), varying only the representation of ψ : a global parametric ψ with the five β -parameters shared across all tickers, a global neural ψ_{NN} shared across all tickers that replaced the parametric form with the neural architecture introduced in the methods, and a sector-specific neural $\psi_{\text{NN}}^{(c)}$ with one network per sector. This design isolated the contribution of model capacity (parametric $\rightarrow$ NN) from the contribution of sharing granularity (global $\rightarrow$ sector). The three-way comparison revealed that both axes mattered and that their contributions were roughly additive (Table 2): moving from the global parametric ψ to the global neural ψ reduced overall RMSE from 9.36% to 8.03% IV (a 14.2% relative improvement) by accommodating smile shapes the log-polynomial form could not express, and moving from the global to the sector-specific ψ_{NN} further reduced RMSE to 6.76% (a 27.8% cumulative relative improvement over the parametric baseline) by letting within-sector shapes specialize away from the global average.

Table 1: Sector partition of the 31-ticker ladder. Each ticker appeared on at least one of the two capture days (2026-04-14 and 2026-04-15); per-sector observation counts were pooled across days. ETF observation counts were substantially larger than other sectors because SPY, QQQ, and IWM carried far deeper strike ladders than individual-name chains.

Sector	Observations	Tickers
ETF	17,533	IWM, QQQ, SPY
Technology	7,689	AAPL, AMD, AVGO, GOOG, INTC, META, MSFT, MU, NVDA, QCOM
Healthcare	3,820	ABBV, AMGN, BMY, JNJ, LLY, MRNA, PFE, UNH
Financials	3,496	BAC, GS, JPM, WFC
Retail	1,989	TGT, UPS, WMT
Energy	1,434	CVX, OXY, XOM

Table 2: Three-way overall RMSE comparison on the 31-ticker ladder (35,961 observations). The parametric baseline fit a single global set of five β parameters; the shared NN replaced the parametric ψ with one global network; the sector NN used one network per sector. All three models fitted one θ_{ticker} per ticker jointly with the shape parameters.

Model	Overall RMSE (% IV)	Relative improvement vs. parametric
Parametric (5 β , global)	9.36	(baseline)
Neural ψ (1 network, global)	8.03	14.2%
Neural ψ (6 networks, sector)	6.76	27.8%

The per-sector decomposition exposed where each incremental model capacity helped most (Table 3). The global NN uniformly improved every sector over the parametric baseline, with the largest absolute gains in retail (8.62% $\rightarrow$ 6.94%) and ETF (8.93% $\rightarrow$ 6.89%), where the parametric log-polynomial struggled to fit shallow ETF smiles and idiosyncratic retail patterns simultaneously. Moving to sector-specific networks delivered a further 1 to 2% IV reduction in every sector except technology, which remained the hardest fit (9.16% with sector NN). Inspection of per-ticker residuals (Table 4) traced this to heterogeneity within the sector itself rather than to a failure of the sector NN as a whole: within technology, names such as AMD (5.80%), META (6.75%), and GOOG (6.80%) fit comfortably under the shared Tech surface, while INTC (12.47%), AVGO (12.59%), MU (10.51%), and MSFT (9.76%) pulled the sector RMSE upward. The pattern was consistent with these outlier tickers carrying idiosyncratic smile shapes (elevated wings, asymmetric skew) that a shared Tech ψ could not simultaneously accommodate without degrading the fit of the better-behaved names. The same story repeated in healthcare, where LLY (5.34%) fit well but PFE (13.33%), ABBV (12.18%), and JNJ (11.79%) drove the sector average. The per-ticker θ_{ticker} values recovered a sensible IV level ranking: high-variance names with concentrated risk (INTC, MU, MRNA, AMD) occupied the top of the sorted order at 60 to 90% baseline IV, large-cap diversified names (AAPL, JPM, WMT) clustered near the middle at 37 to 40%, and the broad-market ETFs (SPY, QQQ, IWM) sat at the low-variance end near 32 to 37%. Within-sector clustering was

evident in the sorted level estimates, confirming that sector membership captured a dominant axis of the cross-sectional IV structure.

Table 3: Per-sector RMSE (% IV) across the three calibration strategies on the 31-ticker ladder. The sector NN reduced RMSE in every sector over the shared NN, with the largest absolute gains in sectors where the global shape poorly represented the local geometry.

Sector	Parametric	Shared NN	Sector NN	Obs. count
ETF	8.93	6.89	5.09	17,533
Financials	6.85	6.44	5.54	3,496
Energy	7.66	6.44	5.97	1,434
Retail	8.62	6.94	6.66	1,989
Healthcare	10.78	9.83	8.83	3,820
Technology	11.65	11.23	9.16	7,689

Table 4: Best and worst individual ticker fits under the sector-specific NN calibration, ranked by RMSE across all 31 tickers. Small-sample tickers (PFE, BMY) showed the expected small-sample penalty; the remaining worst fits were concentrated in technology and healthcare where within-sector shape heterogeneity was largest.

Rank	Ticker	Obs. count	RMSE (% IV)	Sector
<i>Best 5</i>				
1	GS	1,755	4.32	Financials
2	WMT	676	4.71	Retail
3	QQQ	6,386	5.06	ETF
3	SPY	7,748	5.06	ETF
5	XOM	498	5.09	Energy
<i>Worst 5</i>				
27	JNJ	408	11.79	Healthcare
28	ABBV	433	12.18	Healthcare
29	INTC	531	12.47	Technology
30	AVGO	594	12.59	Technology
31	PFE	101	13.33	Healthcare

Beyond the aggregate RMSE comparison, the residual structure under the sector NN was concentrated in the wings (moneyness below 0.90 and above 1.10) rather than near the money, consistent with the in-sample fits of the single-ticker baselines and with the known difficulty of pricing deep-OTM contracts where liquidity is thin and quoted IV is noisier. The learned ψ surfaces were smooth in both $\ln \tau$ and $\ln m$ and displayed the expected qualitative features, including put skew (ψ rising for moneyness below unity), smile curvature in the wings, and a mild term-structure gradient. The sector-to-sector differences in these surfaces motivated relaxing the global-shape assumption in the first place: the healthcare and technology surfaces curved more sharply in the near-term wings than the ETF surface, and the energy surface displayed a subtle asymmetry consistent with the commodity-linked risk profile of the sector. This calibration established the sector-specific neural ψ as the operating point of the framework on the two-day in-sample corpus, but the multiplicative decomposition supports a finer relaxation: one ψ_{NN} per ticker. We took up that extension on the eight-day corpus described below, where per-name sample sizes were large enough to fit per-ticker networks of the same architecture on a meaningful fraction of the universe.

4.3 Per-ticker neural calibration on the eight-day corpus

The eight-day pooled corpus (130,730 filtered observations, 2026-04-14 through 04-24) raised per-ticker sample sizes by roughly an order of magnitude over the two-day in-sample fit, with eighteen of the thirty-one tickers clearing a 2,000-observation threshold sufficient to fit a ψ_{NN} of the same architectural family used at the sector level. We trained one network per qualified ticker on the full pooled corpus, holding all unqualified tickers on the sector ψ_{NN} as a fallback so that every observation was priced under one or the other model. Network width followed sample size: tickers

with at least 5,000 observations received the same $2 \rightarrow 16 \rightarrow 16 \rightarrow 1$ architecture used for the sector networks, while smaller- N qualified tickers used a $2 \rightarrow 8 \rightarrow 8 \rightarrow 1$ network to keep the observation-to-parameter ratio above 20. The per-ticker ψ_{NN} reduced the pooled in-sample RMSE from 9.16% under the sector NN to 8.83% under the combined model (a 3.6% relative reduction) and improved fifteen of the eighteen qualified tickers, with the largest absolute gains concentrated on the data-rich technology names: AAPL (9.36 $\rightarrow$ 7.54%, -1.82%), MU (13.07 $\rightarrow$ 11.65%, -1.42%), NVDA (9.05 $\rightarrow$ 7.81%, -1.24%), MSFT (12.32 $\rightarrow$ 11.54%, -0.78%), and META (10.98 $\rightarrow$ 10.27%, -0.71%). The improvement was visible at the level of the smile fit itself (Figure 1): on NVDA, the per-ticker ψ_{NN} tracked the steep put-skew gradient that the shared Tech surface flattened by averaging across MSFT and AVGO; on MSFT, the per-ticker network captured the gentler curvature near the money that the sector surface overshot; on LLY, the per-ticker fit followed the upward right wing that the Healthcare surface underweighted because of the shallower wings of the larger-cap healthcare names in the pool.

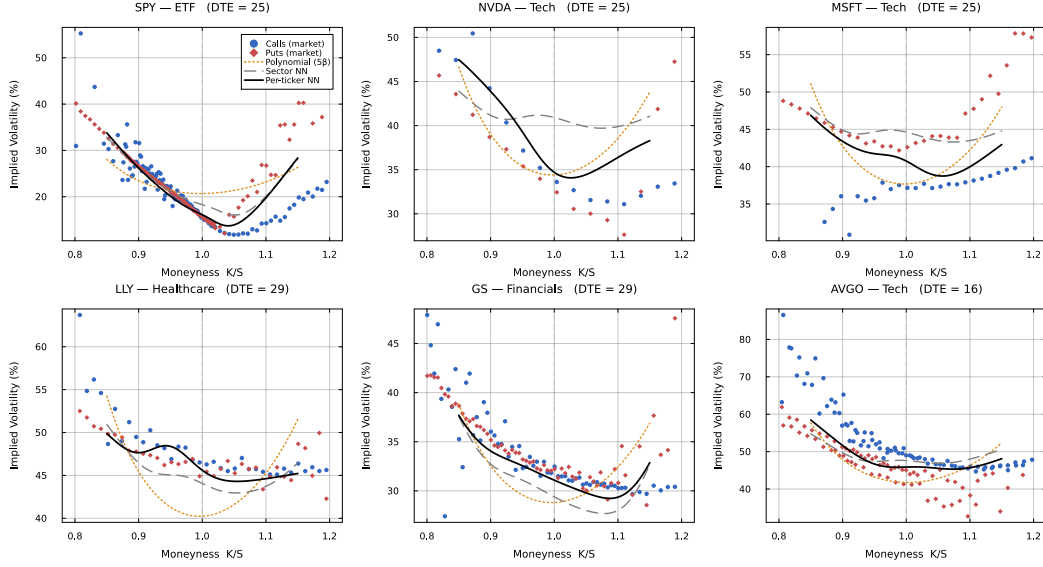


Figure 1: Per-ticker ψ_{NN} smile fits on the eight-day pooled corpus. Representative IV smile panels at mid-DTE for six qualified tickers spanning four sectors (SPY, NVDA, MSFT, LLY, GS, AVGO). The per-ticker network (solid black) tracked the smile shape more tightly than the sector network (dashed gray) on every panel, with the largest gap on NVDA and MSFT where the shared Tech surface flattened to accommodate AVGO’s elevated wings.

The IV-RMSE story translated directly into a dollar-pricing story. Pricing the same ladder contracts via 200-step CRR American at the IV implied by each calibration tier and comparing to the published market mid (Figure 2), the parametric ψ produced systematic dollar errors of \$2 to \$8 across the moneyness range on SPY, NVDA, and LLY, the sector NN halved that error in the near-money region but retained the wing biases, and the per-ticker NN brought the dollar error to within \$1 of the market mid across the full $K/S \in [0.85, 1.15]$ range on all three names. The market-IV reference curve sat at zero by construction and traced the residual CRR-vs-market discretization noise; the per-ticker NN reached this floor on SPY and LLY and stayed within $\sim \$1$ of it on NVDA. The dollar-error figure made the practical implication of the calibration tiers concrete: the difference between sector and per-ticker ψ on a single 30-delta contract was already on the order of the bid-ask spread for these names.

The qualified set was structurally biased toward the names where per-ticker networks could exploit additional capacity. The unqualified set, which retained the sector ψ_{NN} , was dominated by exactly the data-thin names that drove the within-sector RMSE tail of the in-sample fit: PFE (717 observations), BMY (1,018), JNJ (1,304), BAC (1,415), CVX (1,505), OXY (1,495), ABBV (1,516), INTC (1,628), XOM (1,698), MRNA (1,717), WFC (1,864), AMGN (1,977), and QCOM (1,996) all sat below the 2,000-observation cutoff. This pattern set up the sample-size connection between per-ticker capacity and the temporal-generalization story below: the names that benefited least from per-ticker ψ_{NN} were the same names that the sector NN priced worst on the holdout, and both short-

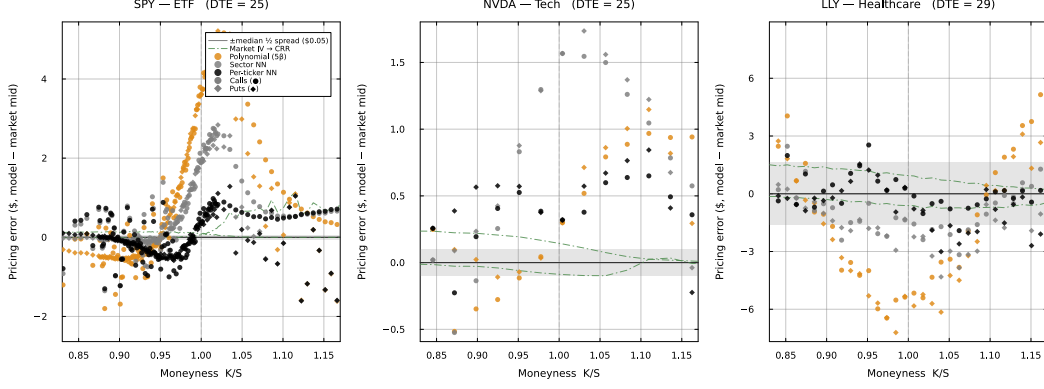


Figure 2: Model-vs-market dollar pricing error (CRR American at the calibrated IV minus market mid) versus moneyiness for SPY, NVDA, and LLY at one representative DTE per ticker. Four model tiers are overlaid on each panel: parametric ψ (light), sector ψ_{NN} (medium), per-ticker ψ_{NN} (dark), and market-IV reference (gray, by construction near zero). The per-ticker fit collapsed the dollar error to within \$1 of the market mid across the tradeable moneyiness range on all three names.

comings were upstream consequences of the same constraint, namely that small-cap tickers and tickers with sparse strike coverage did not yet have the per-name sample size to support either a temporally-validated per-ticker fit or a full per-ticker tail mechanism. We pursued the temporal-generalization question on the same eight-day corpus next.

4.4 Temporal holdout and earnings-aware calibration

The two-day in-sample fit established that the sector-specific ψ_{NN} could absorb the cross-sectional smile geometry of the 31-ticker universe, but it left the temporal-generalization question unanswered: would a ψ trained on one block of capture days price contracts on a held-out subsequent block? We extended the corpus to eight capture days (2026-04-14 through 04-17 and 04-21 through 04-24, comprising 130,730 filtered observations) and split it into a six-day training block (04-14 through 04-22, 103,897 observations) and a two-day temporal holdout (04-23 and 04-24, 26,833 observations). The same 31 tickers appeared on every capture day, so the per-ticker θ_i learned on the training block transferred directly to the holdout. We then asked a sharper question: which fraction of any test-set generalization gap was attributable to scheduled corporate events embedded in the holdout window rather than to architectural overfit? To separate these contributions, we ran the sector ψ_{NN} under three configurations: (A) a pooled control identical to the in-sample architecture above (two inputs $\ln \tau$, $\ln m$, no exclusion); (B) a non-earnings baseline that excluded all training and test rows for which the row's ticker or any same-sector peer was within ± 3 days of an earnings print; and (C) the earnings-aware model introduced in the methods that fed ψ four inputs ($\ln \tau$, $\ln m$, e , e^{peer}) and used the same train and test rows as the control. The peer feature carried sector-coupling information directly: for any ticker i on day t , e^{peer} collapsed to zero if any same-sector equity printed earnings that day, regardless of i 's own distance to its next print.

Table 5: Temporal-holdout sector-NN comparison across the three configurations. The pooled control (A) reproduced the existing four-day reference number (13.03% test RMSE on the same train/test split). Excluding earnings-window observations from both train and test (B) reduced the test RMSE to 7.96% and collapsed the generalization gap from +4.99 to -0.06% IV. Adding the earnings indicator and peer-coupling feature to the input vector (C) on the original train and test rows reduced the pooled test RMSE to 11.98% and the generalization gap to +4.36%.

Configuration	N_{train}	N_{test}	Train RMSE (%)	Test RMSE (%)	Gen. gap (%)
A: pooled control (2 inputs)	103,897	26,833	8.05	13.03	+4.99
B: non-earnings baseline (2 inputs)	29,383	4,516	8.02	7.96	-0.06
C: earnings-aware (4 inputs)	103,897	26,833	7.61	11.98	+4.36

The three-way comparison resolved the generalization question (Table 5). Configuration B revealed that scheduled earnings events accounted for the entire test-time generalization gap of the sector NN: with all earnings-window observations removed from both splits, the test RMSE (7.96%) fell to within rounding of the train RMSE (8.02%), giving a generalization gap of -0.06% IV. The architecture itself generalized faithfully out of sample; what failed was the model’s ability to price options on days when an earnings catalyst dominated the IV surface. Configuration C confirmed that adding event-aware inputs to the shape network recovered roughly one-fifth of the lost performance: relative to the pooled control, the test RMSE dropped from 13.03% to 11.98% (1.05% IV absolute, 8.1% relative), and the generalization gap narrowed from $+4.99$ to $+4.36\%$. The peer-coupling feature did the heavy lifting in this reduction: the per-ticker breakdown showed the largest improvements on Tech names whose own earnings prints fell outside the immediate holdout window but whose sector peers printed during it (Table 6). QCOM (own print 6 days after the holdout) and AMD (12 days after) both benefited from the peer-coupling signal because INTC printed inside the window and dragged their IV surfaces with a $\sim 4\%$ IV reduction in test error each. The same mechanism reduced META, MSFT, and GOOG test RMSE by 2 to 4% IV apiece. INTC itself, the print-day ticker, was the one Tech name whose RMSE rose under the earnings-aware fit ($33.17 \rightarrow 35.94\%$); its out-sized $+2192\%$ EPS surprise produced an unanticipated post-print IV crush that no scheduled-event indicator could predict from the calendar alone.

Table 6: Per-ticker test RMSE on the Tech sector across the pooled control (A) and the earnings-aware fit (C). Columns e and e^{peer} report the median signed days-to-earnings and the minimum same-sector peer distance for each ticker on the holdout days. The largest gains accrued to tickers whose peers printed within the holdout (QCOM, AMD, META, MSFT, GOOG); INTC, the print-day ticker, was the one name that worsened, reflecting an unanticipated outcome that no calendar-based indicator could absorb.

Ticker	e	e^{peer}	A: pooled (%)	C: earnings-aware (%)	Δ (%)
QCOM	+5	1	41.66	37.59	-4.07
AMD	+11	1	40.07	36.50	-3.57
META	+5	1	14.32	10.73	-3.59
MSFT	+5	1	14.00	10.73	-3.28
GOOG	+5	1	9.87	7.90	-1.97
MU	-30	1	16.95	15.59	-1.36
AAPL	+6	1	10.42	10.34	-0.08
AVGO	+30	1	12.58	12.83	+0.24
NVDA	+26	1	9.43	10.47	+1.05
INTC	-1	5	33.17	35.94	+2.77

These results set the operating boundary for what a calendar-driven event indicator can and cannot do. The boundary was sharp: the indicator captured the anticipatory components of event-day IV behavior (pre-earnings expansion on the printer itself, sympathy on same-sector peers) which were a known function of the calendar, but it carried no information about the post-event direction or magnitude when the realization deviated from consensus. The non-earnings baseline of Configuration B is therefore the right reference point against which to compare future event mechanisms: any extension of the model that aims to price event-day IV surfaces must be evaluated against the 7.96% non-event RMSE rather than against the pooled 13.03%, because that gap quantifies the residual event signal that the current architecture cannot reach. We return to this distinction in the discussion, where the failure of the indicator on INTC and its success on the sympathy peers motivate a per-ticker tail mechanism reserved for future work.

5 Forward-simulated short-premium scenarios

To exercise the calibrated framework as a synthetic-data generator and stress test, we forward-simulated 1,000 LLY price paths over a 31 trading-day horizon and tracked the path-conditional fair value of two real LLY option contracts pulled directly from the 2026-04-28 ladder capture: a short put at $K_p = \$825$ (market $\Delta = -0.303$, mid $\$23.30$, market IV 44.4%) and a short call at $K_c = \$940$ (market $\Delta = +0.309$, mid $\$20.76$, market IV 44.0%), both expiring 2026-05-29 with the underlying spot at $S_0 = \$873.83$. Each path drew from the JumpHMM marginal pre-

trained on LLY returns. Conditional on the path, we evolved a Cox–Ingersoll–Ross variance level v_t with mean-reversion $\kappa = 2.0$, vol-of-vol $\sigma_v = 0.5$, and leverage coupling $\rho = -0.6$, and at every trading day along every path we re-priced both contracts via 201-step Leisen–Reimer American with $\sigma_t^2 = v_t \cdot \psi_{\text{NN}}((K/S_t)_{\text{std}}, (T - t)_{\text{std}})$, the per-ticker neural smile/term-structure surface from the ladder calibration. We adopted Leisen–Reimer over the standard CRR lattice because the Peizer–Pratt construction places the strike K at a fixed lattice position by design at every (S, σ) configuration, which gave smooth Greeks via finite differences without the staircase aliasing that CRR’s recombining tree produced when bumps moved K across node boundaries. The seller received the market mid as the entry premium, and the model marked the position to its Leisen–Reimer fair value at every subsequent step; we assumed no scheduled earnings event during the 31-day window so the only sources of variation were diffusion, jumps, and the variance dynamics themselves. The model’s $t = 0$ fair value sat slightly above the market mid on both legs (put: \$25.91 vs. \$23.30, edge +\$2.61; call: \$21.34 vs. \$20.76, edge +\$0.58), so the simulated short was selling at a small discount to the model and would have realised that entry edge on top of the path-conditional terminal P&L.

Figures 3 and 4 render the share-price ensemble alongside the short-put and short-call premium ensembles. On the worst-5% paths (red, Fig. 3), LLY drifted toward and through K_p , the short-put premium more than tripled to roughly \$80, and the position closed with a mean P&L of $-\$58.01$ on those paths. On the top 1–5% paths (green, Fig. 4; the wildest 1% are excluded for legibility), LLY pushed past K_c and the short-call premium tripled despite the calendar pulling extrinsic value toward zero; mean P&L on the top-5% paths was $-\$142.81$, more than $2.4\times$ the mean loss on the worst put paths. On both panels the median path stayed inside the 25–75% band and closed flat at the premium received; the short-premium trade made money on the typical path, with both legs carrying a long left tail.

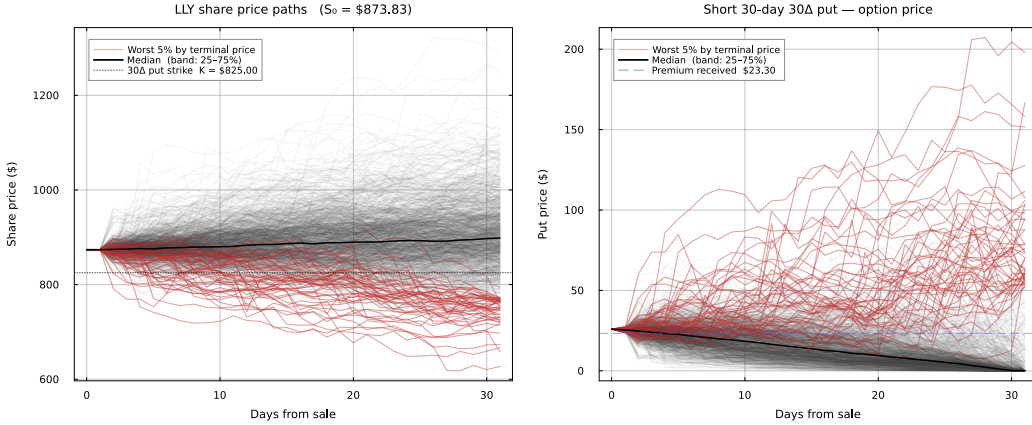


Figure 3: Forward LLY share-price paths (left) and short put premium paths (right) for the real LLY 2026-05-29 \$825 contract, 1,000 simulated trajectories over 31 days. Red: worst 5% by terminal price. Black solid: median; shaded band: 25–75% IQR. Dotted line on the left marks the strike $K_p = \$825$; blue dashed line on the right marks the market mid received at sale, \$23.30.

The IV trajectories in Figure 5 make explicit why the short-call leg lost asymmetrically. Both panels showed the implied volatility at the contract strike rising as $t \rightarrow T$, but the rise was not a model artefact: it is the empirical short-dated wing-bow that the per-ticker ψ_{NN} learned from the ladder corpus. As $(T - t)$ shrank at fixed $\log(K/S)$, the standardised moneyness drifted deeper into the wing where ψ_{NN} is monotonically larger, and a Gaussian diffusion would underprice that wing without the inflation, since short-horizon returns are far more leptokurtic than the central-limit-theorem regime of long-horizon returns. The level v_t contributed only a slow mean-reverting drift around $\bar{\theta}$, so the rise was essentially a smile-shape effect at a fixed point on the surface. On the green tail S_t pushed away from the call strike, dragging the standardised moneyness even further into the wing and amplifying the rise; on the red tail the leverage coupling lifted v_t at the same time as S_t fell, compounding the IV climb.

Figure 6 decomposes the position risk into Δ , Γ , and Vega along the same paths, computed by central finite differences on the same 201-step Leisen–Reimer American pricer used for the base marks (spot bumps of $\pm 1.5\%$ for Δ and Γ ; σ bumps of ± 0.005 for Vega-per-1%-IV). The long-put Δ began near

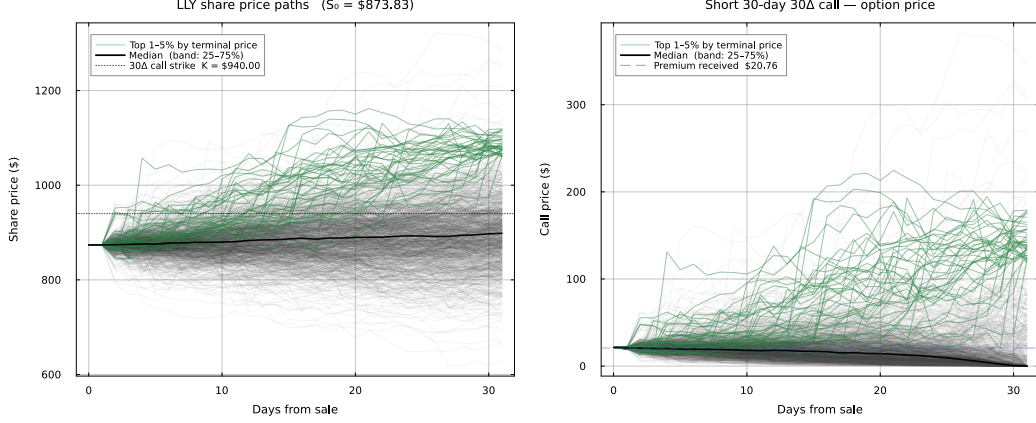


Figure 4: Forward LLY share-price paths (left) and short call premium paths (right) for the real LLY 2026-05-29 \$940 contract. Green: top 1–5% by terminal price; the most extreme 1% are omitted from the overlay so the figure is not dominated by spike outliers, but the underlying summary statistics use the full top 5%. Dotted line marks the strike $K_c = \$940$; blue dashed line marks the market mid, \$20.76.

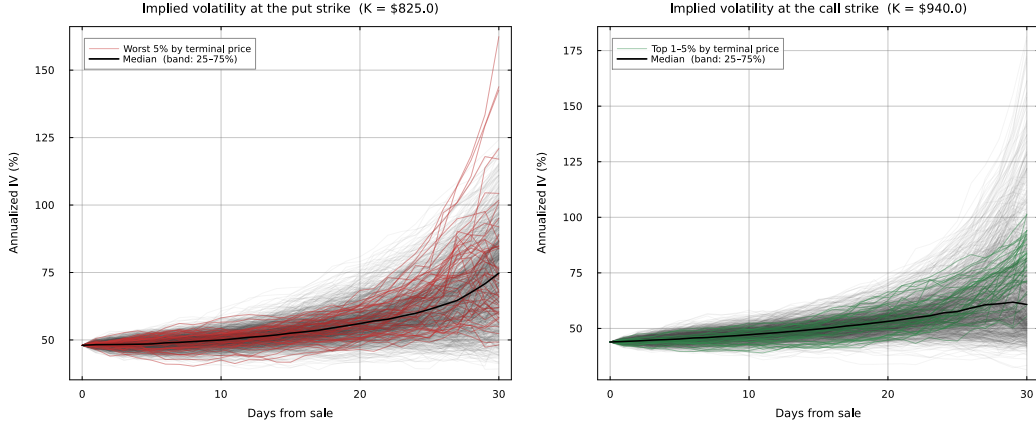


Figure 5: Path-conditional implied volatility at the put strike (left) and call strike (right) over the 31-day horizon. Both panels showed IV rising into expiry: at fixed $\log(K/S)$, shrinking $(T-t)$ pushed the standardised moneyness deeper into the wing of ψ_{NN} , which the per-ticker network learned to be progressively higher and steeper at short horizons. Tail-binned paths (red, green) showed an additional rise driven by S_t moving away from K and, on the put side, leverage coupling lifting v_t on down-paths.

-0.30 and bifurcated: on the worst-5% paths it ran to -1 as the put pinned ITM at expiry, while on the rest it ran to 0 ; the long-call mirror image held at the call strike on the top tail. Γ collapsed to zero on every path that drifted away from its strike but spiked sharply on paths that pinned to the strike near expiry, exhibiting the canonical short-gamma blowup geometry that motivates the practitioner aversion to short-premium expiry-day exposure. Vega decayed monotonically toward zero on every path (the $\sqrt{T-t}$ factor dominates), but the worst-5% put paths and top-5% call paths held materially more vega than the median through about $t = 20$, which exposed the short-premium trader to vol-of-vol risk on precisely the paths that were already losing on Δ . The signs of the position-level Greeks for the short trader are the negatives of the long Greeks shown.

Figure 7 summarises the terminal P&L distributions and Table 7 reports the moments and tail statistics. Both legs showed the characteristic short-premium asymmetry: a thin probability mass piled at the premium ceiling, a long left tail. The short put kept its full \$23.30 premium on 83.1% of paths and closed with a mean P&L of +\$16.94; the short call kept its full \$20.76 premium on 70.2% of paths but its mean P&L was essentially zero (+\$1.86) and the worst-case loss reached

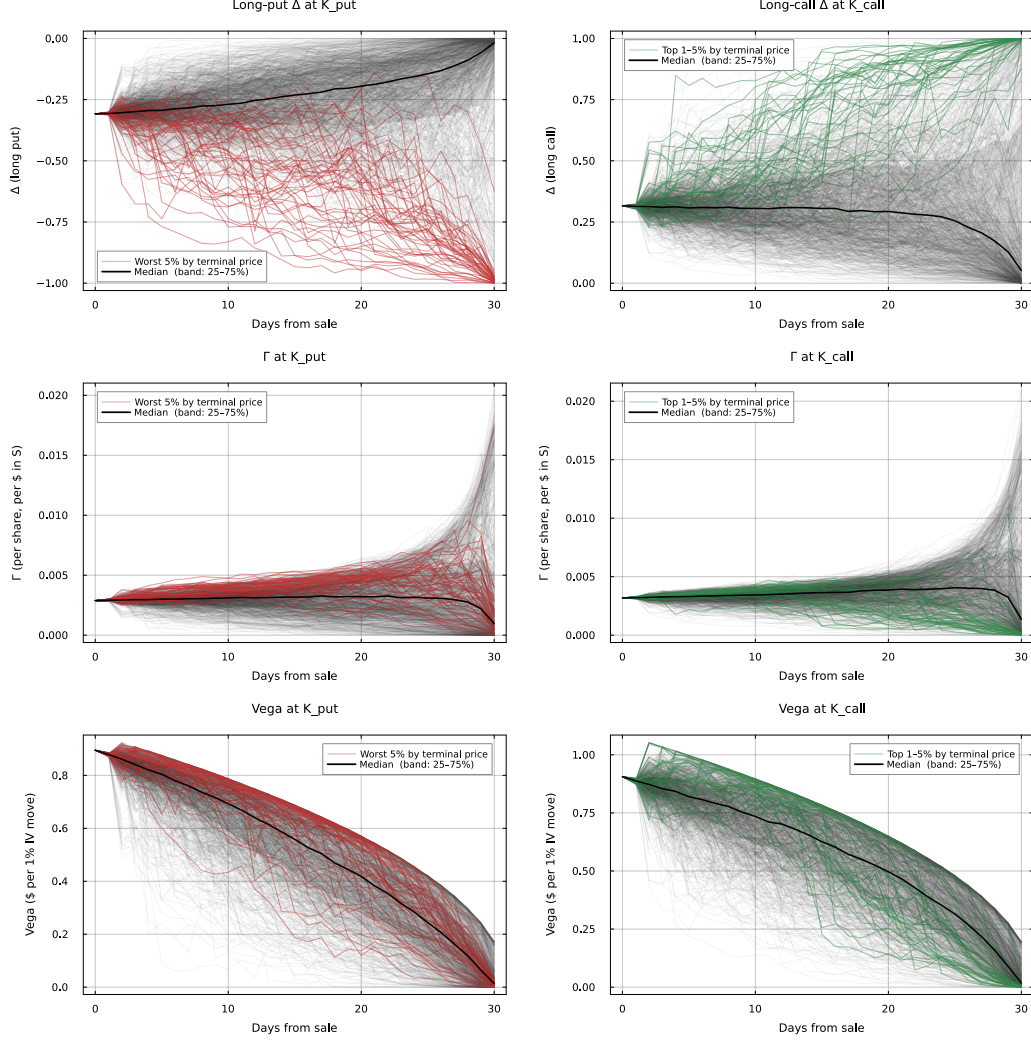


Figure 6: Leisen–Reimer American Greeks (Δ , Γ , Vega) at the put and call strikes over the 31-day horizon, computed by central finite differences on the 201-step Leisen–Reimer pricer at the per-path NN-IV ($h_S = 1.5\%S_t$ for Δ/Γ , $d\sigma = \pm 0.005$ for Vega). Long-Greek convention: short-position Greeks are the negatives. Top row: Δ . Middle: Γ , exhibiting the short-gamma spike on paths that pinned to the strike near expiry. Bottom: Vega, decaying toward zero with $\sqrt{T-t}$ on every path. Tail overlays as in Figs. 3 and 4.

−\$325.62, $1.87\times$ the worst put loss and $15.7\times$ the premium itself. The asymmetry between the two legs reflected the negative skew that LLY’s per-ticker ψ_{NN} encodes (higher IV at down-strikes than at equally-far up-strikes) together with the leverage coupling that lifted v_t on down-paths; the put leg was implicitly priced for the down-tail and earned a fair premium for it, while the call leg encountered a fatter empirical right-tail than its strike-symmetric BS-implied premium accounted for.

6 Discussion and Future Work

The calibration and scenario results validated the core claim of this study: a modified Heston process with a state-dependent θ -function produced realistic IV surfaces from structural return dynamics alone, and the same decomposition of θ into a per-ticker level and a shape function ψ accommodated a hierarchy of representations spanning a parsimonious analytic form, a globally shared neural surrogate, and a sector-specific neural surrogate. The single-DTE parametric smile captured

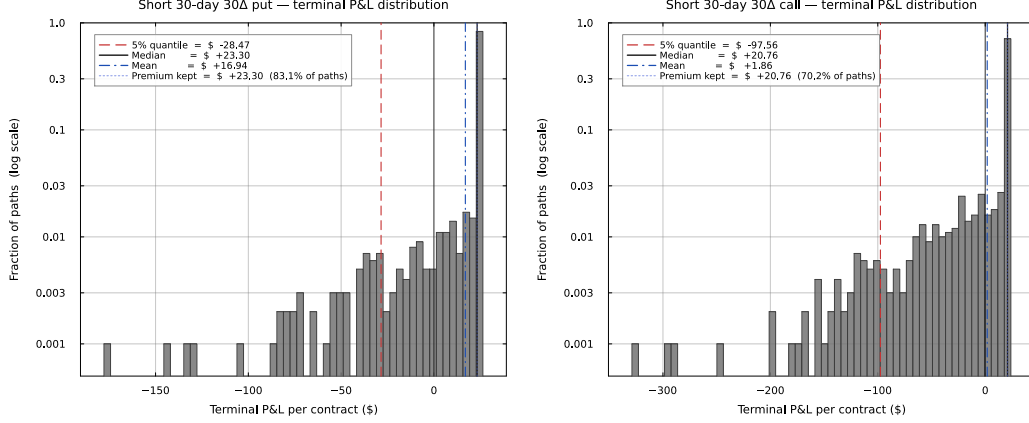


Figure 7: Terminal P&L distributions per contract (1,000 paths, log-y). Both legs showed the short-premium signature: a tall spike of paths keeping the full premium, and a long left tail. The short call’s left tail was substantially fatter than the put’s despite a smaller premium received.

Table 7: Terminal P&L statistics from 1,000 LLY 31-day forward simulations of the real 2026-05-29 \$825 put and \$940 call. Entry premium is the market mid from the 2026-04-28 capture. The model’s $t = 0$ Leisen–Reimer fair value at NN-IV is reported as a reference; statistics are per contract.

Statistic	Short put (\$)	Short call (\$)
Strike K	825	940
Market mid (entry premium)	+23.30	+20.76
Model $t = 0$ fair value	+25.91	+21.34
Entry edge (model – market)	+2.61	+0.58
Mean P&L	+16.94	+1.86
Median P&L	+23.30	+20.76
Std P&L	20.22	42.36
5%-tile P&L	−28.47	−97.56
Worst-case P&L	−174.60	−325.62
Premium kept in full	83.1%	70.2%

the essential features of the NVDA IV smile, and the smile shape generalized across semiconductor names with only a scalar level adjustment; the multi-expiration calibration extended this to the full NVDA IV surface with the five-parameter ψ function and the quadratic DTE term β_5 that captured the U-shaped ATM term structure. The equilibrium initialization $v_0 = \theta(t=0)$ eliminated v_0 as a free parameter and produced the smile, skew, and term structure by construction; this was not merely a computational convenience but a structural constraint that ensured the initial IV surface was fully determined by the θ -function, consistent with the view that IV surfaces reflect the conditional distribution of returns under the risk-neutral measure [Cont and da Fonseca \[2002\]](#). The ψ -function played an analogous role to the SVI parameterization [Gatheral \[2004\]](#) in that it provided a parsimonious representation of the smile and term structure, but with a critical difference: the parameters entered a generative model rather than a static interpolation, so the calibrated surface could evolve dynamically as the underlying HMM state changed.

Scaling the calibration to the 31-ticker two-day ladder revealed the limits of the parametric form as a universal shape function and motivated the hierarchy of neural representations. A single global ψ fit across the six sectors achieved an overall RMSE of 9.36% IV, dominated by systematic misfits in sectors whose smile geometry departed from the semiconductor baseline (healthcare at 10.78%, technology at 11.65%). Replacing the parametric form with a globally shared neural ψ reduced the overall RMSE to 8.03% without changing the sharing granularity, confirming that part of the gap was a functional-form bottleneck in the log-polynomial representation. Loosening the sharing to one network per sector reduced the overall RMSE further to 6.76%, with the sector NN strictly dominating the shared NN in every sector, which established that both axes (capacity and granularity)

contributed independently. The remaining error was concentrated in two structural sources: within-sector heterogeneity in technology and healthcare, where specific tickers (INTC, AVGO, MU, PFE, ABBV) carried idiosyncratic smile shapes not expressible by a shared sector-level ψ , and small-sample effects for tickers with fewer than 200 observations (PFE at 101 observations). Both sources are addressable through the same decomposition: the per-ticker roadmap relaxes shape sharing to the ticker level for the data-rich names, and additional capture days directly reduce small-sample variance for the thin names.

Extending the corpus to eight capture days and running a temporal holdout reframed the generalization question more sharply than the in-sample RMSE alone could. The pooled-control sector NN trained on six days and evaluated on the next two carried a 4.99% IV generalization gap (8.05% train, 13.03% test), which on its face suggested a substantial overfit. Excluding observations within ± 3 days of any same-sector earnings print from both train and test eliminated that gap entirely (8.02% train, 7.96% test, -0.06% generalization gap), revealing that the architecture itself generalized faithfully out of sample and that the entire 4.99% gap was attributable to scheduled corporate events embedded in the holdout window. Adding earnings-distance and same-sector peer-distance features to the shape network closed roughly one fifth of the gap on the original train and test rows (11.98% test, $+4.36\%$ generalization gap). The per-ticker decomposition exposed the mechanism: the peer-coupling feature carried the IV-spillover signal, so QCOM (own print six days after the window) and AMD (twelve days after) saw $\sim 4\%$ IV reductions in test error from the INTC print on a holdout day, and the same mechanism cut META, MSFT, and GOOG test RMSE by 2 to 4% IV. INTC itself was the only Tech name whose error rose under the earnings-aware fit, driven by an outsized $+2192\%$ EPS surprise that the calendar-based indicator could not predict. The boundary was therefore sharp: the indicator captured the anticipatory components of event-day IV behavior (pre-print expansion on the printer, sympathy on same-sector peers), which were a function of the calendar alone, but it carried no information about the post-event realization when consensus was wrong. The non-earnings baseline of 7.96% test RMSE is the right reference for what the current architecture can do under faithful temporal generalization; the residual gap to the pooled 13.03% quantifies the part of the surface that requires a richer event mechanism than the calendar.

Several limitations constrained the scope of the current conclusions and pointed directly to the natural extensions that followed from them. The cross-sector partition used conventional GICS-style groupings; whether alternative groupings (mega-cap versus semiconductor within technology; pharma versus biotech within healthcare) improve the sector-NN fit without fragmenting the corpus is an empirical question that additional data will clarify. The forward scenario analysis demonstrated the framework on real 31-DTE 30-delta LLY contracts pulled from the 04-28-2026 capture and produced a coherent path-conditional risk decomposition (Leisen–Reimer American Greeks via finite differences, terminal short-premium P&L, IV trajectories, and the entry-edge gap between the model’s lattice fair value and the market mid), but the validation was limited to a single ticker, a single 30-day window, and a single market regime; broader validation across sectors and volatility environments, and a re-run of the scenarios with the sector-NN ψ for tickers without per-ticker networks, would strengthen the generalizability claim. The current scenarios also assumed no scheduled earnings event during the simulation window, which the earnings-aware calibration flagged as the dominant residual driver of test-time error; a natural extension is to forward-simulate across an earnings print and quantify how the path-conditional Greek profile and terminal P&L shift when the event-aware ψ feeds the same simulation pipeline. The earnings-aware fit recovered the anticipatory portion of the event-day signal but left the surprise-driven portion uncaptured. A natural extension restored the per-ticker tail term that the simulation pipeline already supports for forward state evolution and that the calibration omitted: replacing the operational θ_i with $\theta_i \cdot (1 + \gamma_i M_i^{(t)})$, where $M_i^{(t)}$ was the JumpHMM tail-state probability for ticker i at time t and γ_i was a learned per-ticker tail sensitivity, would absorb idiosyncratic IV inflation from any source whose return-side imprint already drove the HMM into a tail state, including but not limited to earnings surprises. Restoring this term required state-conditioned histories rather than observation-day surfaces, so it sat naturally at the simulation interface: the calibrated ψ_{NN} remains the IV shape function, the calendar-based earnings inputs (e , e^{peer}) remain the scheduled-event indicator, and $\gamma_i M_i$ becomes the unscheduled tail mechanism populated from the JumpHMM at simulation time. The natural reference point for evaluating either of these extensions is the non-event 7.96% test RMSE established by the earnings-aware calibration, against which any further improvement would represent a genuine reduction in the residual event signal rather than a recovery of cross-sectional fit already accounted for by the

sector ψ . Beyond these specific limitations, the primary intended use of this framework was generating realistic synthetic training data for machine-learning models in finance, with applications ranging from training option price predictors and learning delta-hedging policies via reinforcement learning Buehler et al. [2019] to augmenting scarce real option data for transfer learning. The framework naturally supported computation of Greeks (Δ , Γ , $\mathcal{V}$, Θ) via finite differences on the binomial-tree pricer, and the LLY scenario found that the Leisen–Reimer variant produced smooth Greeks at modest tree depth where CRR’s recombining lattice exhibited node-snapping aliasing in σ and S bumps. Including these Greeks in the synthetic dataset would support hedging applications. Finally, the current mood mechanism created correlated IV dynamics through a shared stress signal; a richer model could introduce ticker-pair IV correlations through the copula structure, producing a full cross-asset IV surface that respected observed correlation patterns between individual-name and index volatility.

7 Conclusion

In this study, we developed a framework for generating synthetic American option prices with self-consistent implied volatility dynamics. The framework linked a Jump Hidden Markov Model to a modified Heston stochastic variance process via a state-dependent mean-reversion target $\theta(t)$, and priced the resulting IV through a CRR binomial tree with early exercise. The equilibrium initialization $v_0 = \theta(t=0)$ eliminated the initial IV surface as a free parameter, and the endogenous market mood signal produced correlated IV spikes during stress events without external data.

We calibrated the smile and term-structure shape function ψ through a hierarchy of representations. A five-parameter parametric form fitted on NVDA options first demonstrated that the smile shape generalized across AMD, MU, and INTC with cross-ticker RMSE of 1 to 5% IV using only a scalar level adjustment per ticker, and a multi-expiration fit across seven expirations revealed a U-shaped ATM term structure captured by the quadratic DTE extension. Scaling to a 31-ticker, two-day, six-sector option ladder exposed the limits of the parametric form as a universal shape function, and we demonstrated that replacing it with a globally shared neural surrogate and then with sector-specific neural surrogates reduced the overall fit error by 14% and 28% respectively, with independent contributions from added model capacity and loosened sharing granularity. Extending to an eight-day corpus and running a temporal holdout further showed that scheduled earnings events accounted for the entire test-time generalization gap of the sector NN, and that adding calendar-derived earnings-distance and same-sector peer-distance features to ψ recovered roughly one fifth of that gap; the residual portion arose from earnings surprises whose direction and magnitude could not be inferred from the calendar alone. We then exercised the calibrated framework as a synthetic-data generator on a real LLY 2026-05-29 \$825 put and \$940 call (both 31 DTE, 30-delta) pulled from the 04-28-2026 capture, forward-simulated 1,000 price paths under the JumpHMM and the per-ticker ψ_{NN} , and tracked path-conditional implied volatility, premium decay, finite-difference Leisen–Reimer Greeks, and terminal short-premium P&L from a single coherent simulation. The short put and short call kept their full premium on 83% and 70% of paths respectively, and the asymmetric tail risk of the short call (worst-case loss $1.87\times$ that of the short put despite a smaller premium received) emerged naturally from the negative skew that the per-ticker ψ_{NN} encoded together with the leverage coupling that lifted v_t on down-paths. The framework was implemented as an open-source Julia package and is available at <https://github.com/varnerlab/heston-implied-volatility-model>.

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A Parametric baseline: additional calibration and diagnostics

The main text reports a single ATM term-structure fit to establish the parametric ψ as a baseline. This supplement reports the two earlier-stage parametric calibrations that preceded the neural surrogate: a single-DTE fit on four semiconductor tickers that demonstrated cross-ticker transfer of the smile shape, and the full multi-expiration NVDA calibration from which the main-text figure was drawn, including the parameter values, residual diagnostics, and IV surface reconstruction.

A.1 Single-DTE four-ticker semiconductor calibration

The empirical starting point for the parametric form was a set of option chains for four semiconductor stocks: NVDA, AMD, MU, and INTC. All chains were single-date snapshots sourced from barchart.com (October to November 2025) with DTEs ranging from 41 to 80 days, filtered to moneyness $K/S \in [0.70, 1.30]$ and $IV < 200\%$ to exclude illiquid deep ITM/OTM contracts with unreliable IV (Table 8). Because each chain was a single-date snapshot, all observations shared one HMM state s_0 and market mood M_0 , so the model IV at a given strike reduced to $\sigma_{\text{model}}(K) = \sqrt{\theta_{\text{base}} \cdot \psi(\text{DTE}, K/S)}$, where $\theta_{\text{base}} = \theta_{s_0} \cdot (1 + \gamma M_0)$ absorbed the state and mood into a single scalar. We fixed $\beta_1 = 0.20$ (term-structure slope) because single-DTE snapshots could not identify it; the remaining parameters $(\theta_{\text{base}}, \beta_2, \beta_3, \beta_4)$ were fitted by minimizing the mean squared IV error on NVDA using Nelder-Mead optimization. We chose NVDA as the calibration ticker because it had the richest chain (142 contracts after filtering).

The calibrated parameters (Table 9) produced the expected smile features: the negative β_2 generated put skew, the positive β_4 generated smile curvature, and the interaction term β_3 captured how the skew flattened at longer maturities. The calibration achieved an RMSE of 1.16% IV on NVDA.

Table 8: Option chain summary for the single-DTE semiconductor dataset. Contract counts are reported after filtering to the tradeable moneyness range.

Ticker	Spot (\$)	DTE	ATM IV (%)	Contracts
NVDA	207.04	80	44.6	142
AMD	225.78	58	59.0	36
MU	237.02	41	72.4	34
INTC	37.78	65	53.6	46

To determine whether the smile shape generalized across tickers, we applied the $(\beta_1, \beta_2, \beta_3, \beta_4)$ calibrated on NVDA to AMD, MU, and INTC, re-optimizing only the scalar θ_{base} per ticker. Four parameters sufficed to capture the smile shape within the tradeable moneyness range, and the cross-ticker validation revealed that the shared smile shape transferred well to AMD (2.34%) and reasonably to MU (3.99%) and INTC (4.65%; Table 10). The higher RMSE for MU and INTC reflected differences in skew steepness across these names; MU exhibited a steeper put skew than NVDA, while INTC had a flatter OTM call wing, suggesting that per-ticker β calibration would further reduce errors when multi-strike data was available. The residual analysis confirmed that the errors were concentrated in the wings ($K/S < 0.85$ and $K/S > 1.15$), with near-ATM residuals below 1% IV for all tickers. These results established that the ψ -function captured the dominant smile features with a small number of parameters, and motivated the multi-expiration extension reported below.

Table 9: Calibrated smile parameters from the single-DTE NVDA fit (DTE = 80).

Parameter	Interpretation	Value
β_1	Term-structure decay (fixed)	0.20
β_2	Skew (asymmetry)	-2.07
β_3	DTE $\times$ skew interaction	0.34
β_4	Smile curvature	1.43
θ_{base}	Variance level (NVDA)	0.084

Table 10: Cross-ticker RMSE (% IV) from the single-DTE calibration. The smile shape $(\beta_2, \beta_3, \beta_4)$ was shared from NVDA; only θ_{base} was re-fitted per ticker.

Ticker	θ_{base} (IV level)	RMSE (% IV)
NVDA (calibration)	28.9%	1.16
AMD	38.7%	2.34
MU	49.0%	3.99
INTC	35.9%	4.65

A.2 Multi-expiration NVDA calibration

Having validated the smile shape across tickers, we addressed the term-structure limitation by collecting NVDA call and put chains at seven expirations on the same observation date (April 6, 2026): 0, 2, 4, 7, 14, 25, and 46 DTE, with NVDA trading near \$176.70. The cross-sectional snapshot provided the multi-expiration data needed to identify β_1 directly and to resolve the full term-structure geometry, yielding 73 observations after filtering to moneyness $K/S \in [0.90, 1.10]$ with positive volume. The six parameters $(\sigma_{\text{base}}, \beta_1, \dots, \beta_5)$ of equation (6) were fitted by minimizing mean squared IV error across all seven expirations simultaneously.

The calibrated coefficients (Table 11) combined a negative β_1 with a positive β_5 to produce the U-shaped ATM curve: at short maturities the linear term dominated (IV decayed rapidly from the 0 DTE spike), while at longer maturities the quadratic term bent the curve upward. The trough occurred near DTE ≈ 8 , where IV reached 32.5%, a $1.38\times$ ratio between the 0 DTE peak and the trough. The skew parameters ($\beta_2 = -8.45$, $\beta_3 = 1.62$) confirmed that the volatility skew steepened at shorter maturities and flattened as DTE increased, consistent with the concentration of crash risk

premium in near-term options Bates [2000]. The overall fit achieved an RMSE of 2.31% IV across all seven expirations and an RMSE of 1.15% IV across the 2 to 46 DTE range reported in the main text. The fit was tightest at intermediate and long maturities (7 to 46 DTE: RMSE < 1.3%), with the largest residuals at 0 DTE (RMSE = 6.57%) where the gamma-compressed smile shape deviated from the log-moneyness functional form. The 0 DTE level bias was eliminated (−0.1%), confirming that the U-shaped term structure was captured correctly even though the extreme smile curvature at expiration remained a source of scatter.

Table 11: Calibrated ψ parameters from the multi-expiration NVDA dataset.

Parameter	Interpretation	Value
σ_{base}	Base ATM IV level	45.0%
β_1	DTE decay	−0.622
β_2	Skew (asymmetry)	−8.448
β_3	DTE $\times$ skew	1.617
β_4	Smile curvature	−1.242
β_5	DTE curvature	0.150

B Holdout diagnostics motivating the earnings-aware mechanism

The temporal-holdout result of the main text attributed the test-time generalization gap to scheduled corporate events rather than to a broad-market regime shift, but that attribution required ruling out the alternative explanation that the 04-23 and 04-24 capture days had simply landed inside a volatility expansion that the training block did not see. Two diagnostics established the alternative as untenable and pinned the blow-up to ticker-level event behavior. First, the broad-market volatility index VIX traced a flat path through the holdout window: VIX closed at comparable levels on the final training day and the two holdout days, with no intraday excursion during the holdout that approached even a one-standard-deviation move relative to the trailing six-day window. A regime-shift hypothesis would have required the index to have moved materially over the holdout, and it did not. Second, the per-ticker ATM IV trace revealed that the IV expansion on 04-23 and 04-24 was concentrated entirely in the technology sector, and within that sector concentrated on the names with earnings prints inside or adjacent to the holdout window: INTC (which printed inside the window), MSFT, META, and GOOG all exhibited ATM IV elevations of 5 to 30 percentage points relative to their training-block baselines, while non-Tech names traced ATM IV paths consistent with their pre-holdout levels. The combination established that the holdout RMSE jump was not a market-wide effect that the architecture had failed to absorb but a localized response to the earnings calendar, and it directly motivated the earnings indicator and peer-coupling feature reported in Configuration C of Table 5.